

Apply Gender Here

Sophia Serice, Ada Totten

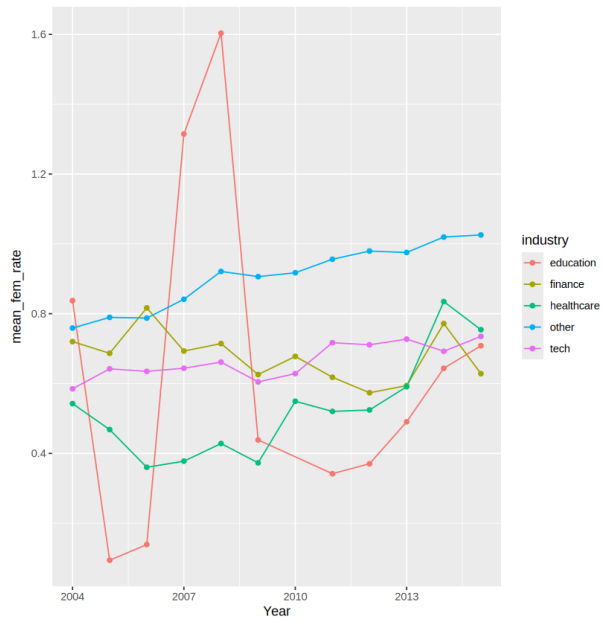
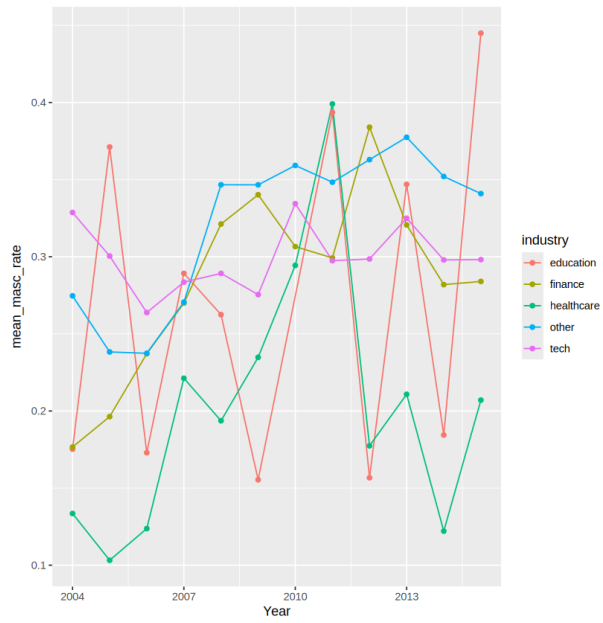
September 27, 2026

Abstract

We hypothesized that there would be some correlation between gendered—based on the APA’s list of gendered terms—language and industry advertisements, and how that sentiment changes over time. Through analysis of job advertisements 2004-2015 (Online Job Postings) and 2023-2024 (LinkedIn Job Postings (2023 - 2024)), pulled from free open source data on Kaggle. Through analysis of this data, we have concluded that there is a relationship between both job industry, gendered language in job advertizements and that this has varied over time. For this paper we used the R programming language in conjunction with the Tidyverse library as we have previous experience using those in a linguistic context. Possible further explorations could examine more historical data.

1 Data





Gendered Word Usage in Job Postings by Industry and Era

Percentage of ads containing at least one gendered word



2 Conclusions

As a result of our analysis, we have concluded that there is apparent gendered wording in job advertisements and the frequency of that wording has changed from 2004-2015 and from 2023-2024. There has been an increase in gendered word frequency in both masculine and feminine words from 2004 to 2024 in all categories, we predict this is because of an increased focus on equality and a diverse workforce when looking for employees. The frequency of gendered words does vary significantly from industry to industry until we reach 2023-24.

3 Production & Code

Loading data from .csv files:

```
library(tidyverse)
options(tidyverse.quiet = TRUE)

data_jobposts <- read_csv("content/data_job_posts.csv")
posts2023_4 <- read_csv("content/postings.csv")
industries <- read_csv("content/industries.csv")
```

Cleaning data:

The data had a substantial amount of extaneous information outside of the scope of this project so we had to excise unnecessary data to make the code more comprehensible.

```
jobposts.clean <- data_jobposts %>%
  select(-c(Duration, StartDate, Audience, Eligibility, Term, AnnouncementCode, Company, IT, Month, Attach, AboutC, Notes, Deadline,
    OpeningDate, ApplicationP, Salary, RequiredQual, JobRequirement))

clean.posts2023 <- posts2023_4 %>%
  select(-c(max_salary, pay_period, location, company_id, views, med_salary, skills_desc, listed_time, posting_domain, sponsored, work_type,
    currency, compensation_type, normalized_salary, zip_code, fips, min_salary, applies, original_listed_time, remote_allowed,
    application_url, application_type, expiry, closed_time, formatted_experience_level, job_posting_url, formatted_work_type))
```

Cleaning data:

We then stored the APA's gendered words in two vectors.

```
masc_words <- c("active", "adventurous", "aggressive", "ambitious", "analytic", "athletic", "autonomous", "boast",
  "challenger", "challenging", "competent", "confident", "courageous", "decide", "decisive", "decision",
  "determination", "dominant", "domination", "force", "forceful", "greedy", "headstrong", "hierarchy",
  "heirarchical", "hostile", "hostility", "impulsive", "independence", "independant", "individual",
  "individualistic", "intellectual", "intellect", "lead", "leader", "leading", "logic", "masculine",
  "objective", "opinion", "opinionated", "outspoken", "persist", "principle", "reckless", "stubborn",
  "superior", "self-confident", "self-confidence", "self-sufficient", "self-sufficiency", "self-reliant",
  "self-reliance")

fem_words <- c("affectionate", "child", "childlike", "childish", "childhood", "cheerful", "cheer", "commit", "committed",
  "communal", "compassion", "compassionate", "connect", "connected", "considerate", "cooperative", "cooperate",
  "dependable", "depend", "dependant", "emotional", "emotion", "empathetic", "empathy", "empath", "feminine",
  "flatterable", "gentle", "honest", "interpersonal", "interpersonally", "interdependent", "interdependence", "interdependently",
  "kind", "kinship", "loyal", "loyalty", "modesty", "modest", "nag", "nurture", "nurturing", "pleasant", "pleasantly",
  "polite", "quiet", "quietly", "quietness", "responsive", "responsiveness", "sensitivity", "sensitive", "sensitiveness",
  "submissive", "support", "supportive", "supporting", "sympathetic", "sympathy", "sympath", "tender", "tenderness",
  "together", "togetherness", "trust", "trusting", "understanding", "understand", "warm", "warmness", "whining", "yield", "yielding")
```

Analysis and plotting:

Lastly, we generated graphs using assorted ggplot functions such as the ones provided (after performing some small operations to make the data more easily parsable via code).

```
jobposts.masc.plot <- ggplot(plot, aes(x = Year, y = mu)) +
  geom_line(color = "orange", size = 1) +
  geom_point(color = "red", size = 2) +
  labs(title = "masculine average over time", x = "Year", y = "masculine word rate") +
  theme_grey()

jobposts.masc.plot

jobposts.clean %>% group_by(Year) %>%
  summarize(mu=mean(fem_rate, na.rm = TRUE)) -> plot2

jobposts.fem.plot <- ggplot(plot2, aes(x = Year, y = mu)) +
  geom_line(color = "blue", size = 1) +
  geom_point(color = "red", size = 2) +
  labs(title = "feminine average over time", x = "Year", y = "feminine word rate") +
  theme_grey()
```